

Formulating the OLS Regression Problem as Error-Restricted NLP to Simultaneously Test Gauss-Markov Conditions

Shailesh J. Divey¹, James J. Cochran²

¹University of Oregon, ²The University of Alabama at Tuscaloosa

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Gauss-Markov Conditions

For a linear model $y = x\beta + \epsilon$

Linearity:

$$E[y|x_1, x_2, \dots, x_k] = E[y|x] = x'\beta$$

Conditional Mean:

$$E[\epsilon|x_1, x_2, \dots, x_k] = 0$$

Homoskedasticity:

$$Var[\epsilon|x_1, x_2, \dots, x_k] = Var[\epsilon] = \sigma^2$$

Autocorrelation:

$$Var[\epsilon] = \sigma^2 I$$



Minimizing Sum of Squared Errors

Problem Formulation

$$\text{minimize } \sum_i f_i(\epsilon_i) = \epsilon_i^2 = (Y_i - (\beta X_i))^2$$

s. t.

$$p_{\text{Independence}} \geq p_{\text{Independence}}^*$$

$$p_{\text{Normality}} \geq p_{\text{Normality}}^*$$

$$p_{\text{Fit}} \geq p_{\text{Fit}}^*$$

$$p_{\text{Homoskedasticity}} \geq p_{\text{Homoskedasticity}}^*$$

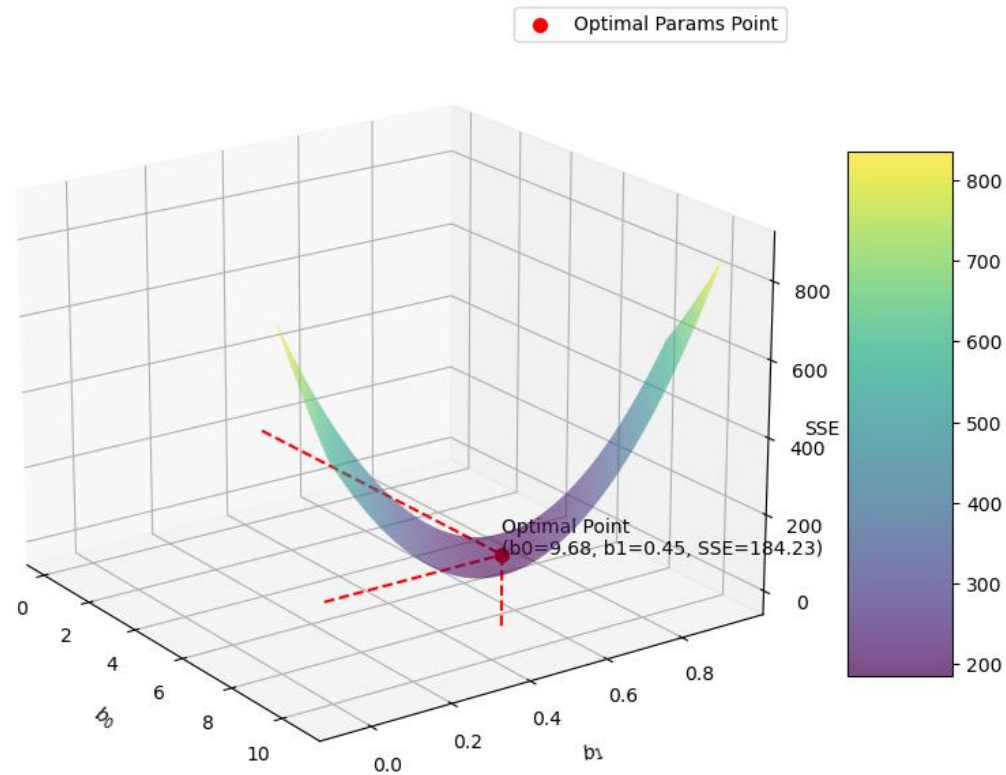
$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta} \end{pmatrix} = \arg \min (||y - (\beta_0 + X\beta)||_2)$$

s. t.

$G - M$ constraints

Independence

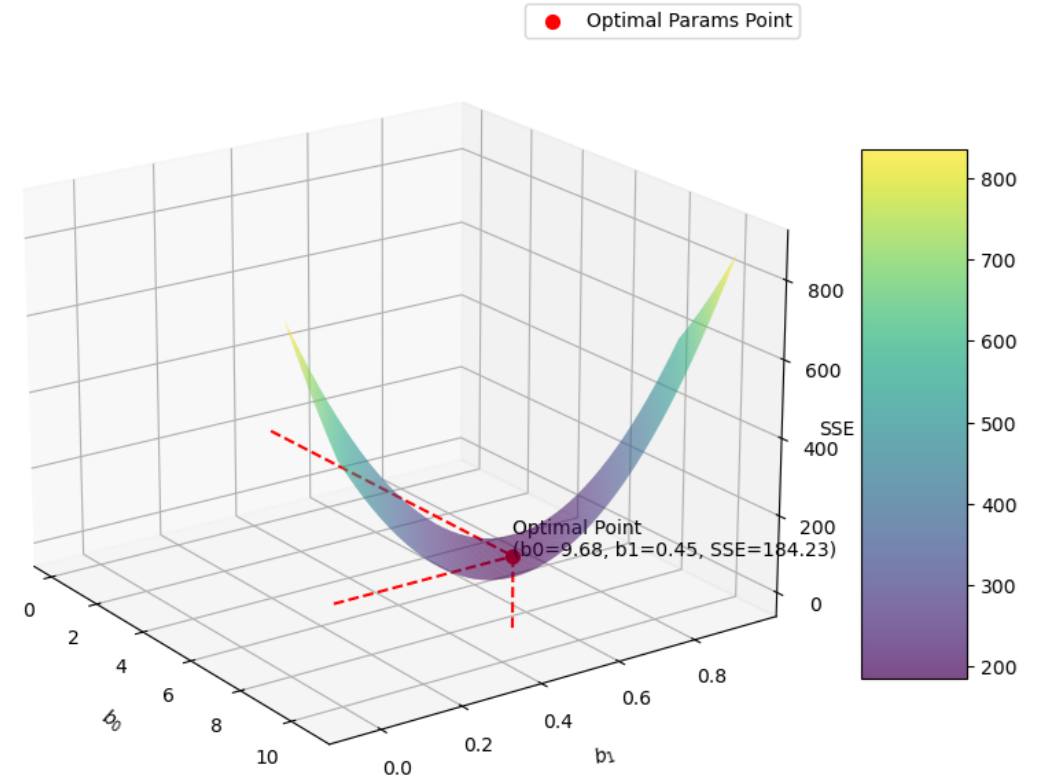
SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Independence Testing



$p\text{-val}_{\text{start}}=0.11$; $p\text{-val}_{\text{end}}=0.26$

Normality

SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Normality Testing

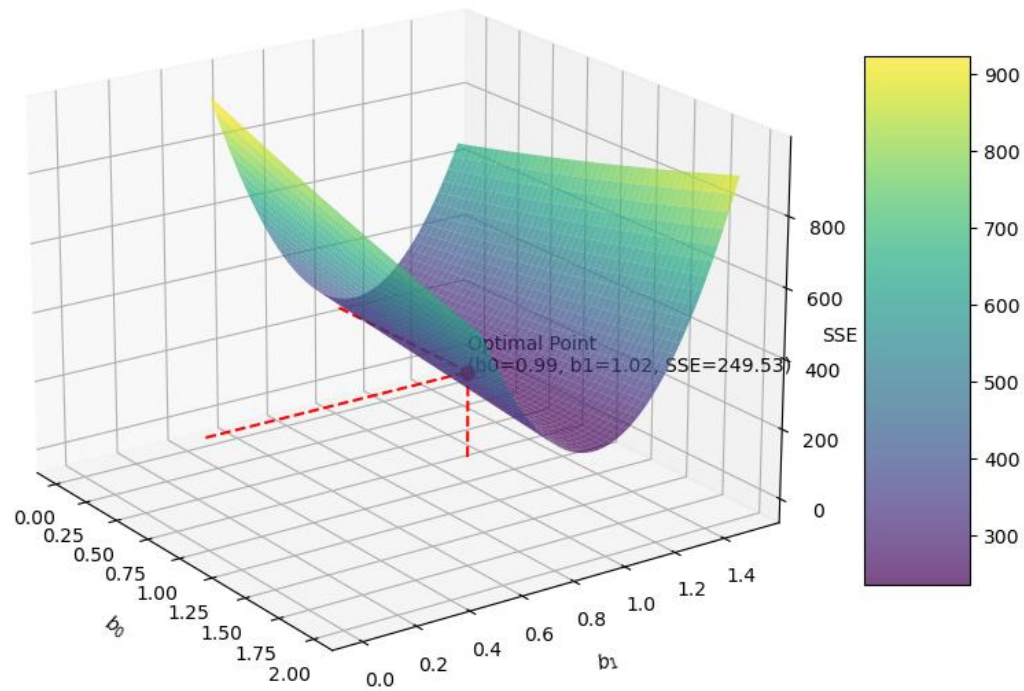


$p\text{-val}_{\text{start}}=0.07$; $p\text{-val}_{\text{end}}=0.94$

Fit/Linearity

SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Linearity Testing

● Optimal Params Point

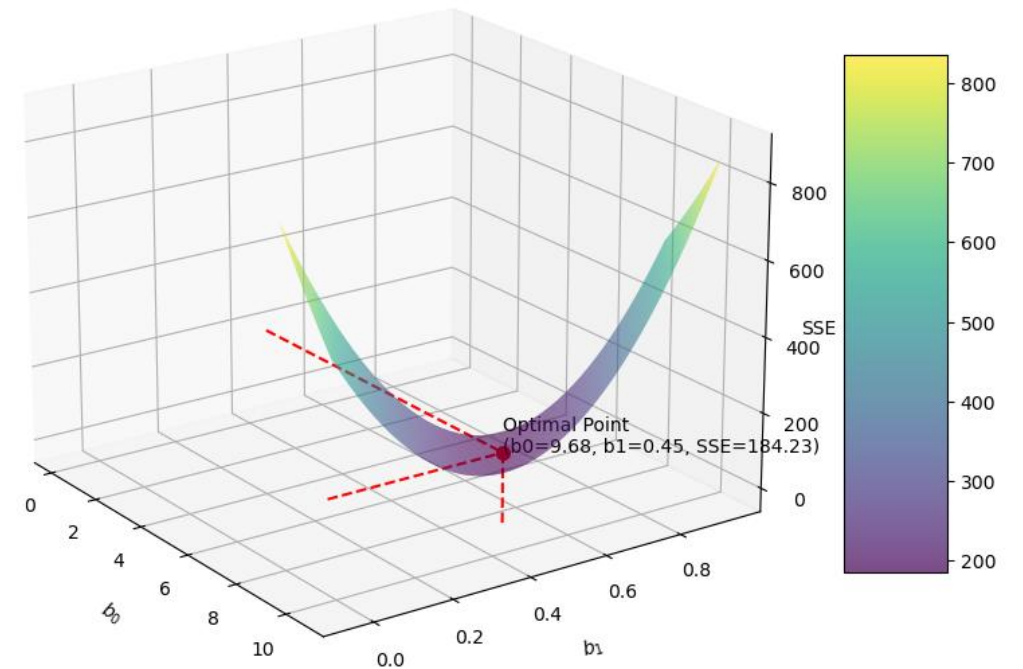


$p\text{-val}_{\text{start}}=0.99; p\text{-val}_{\text{end}}=0.97$

Homoscedasticity

SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Homoscedasticity Testing

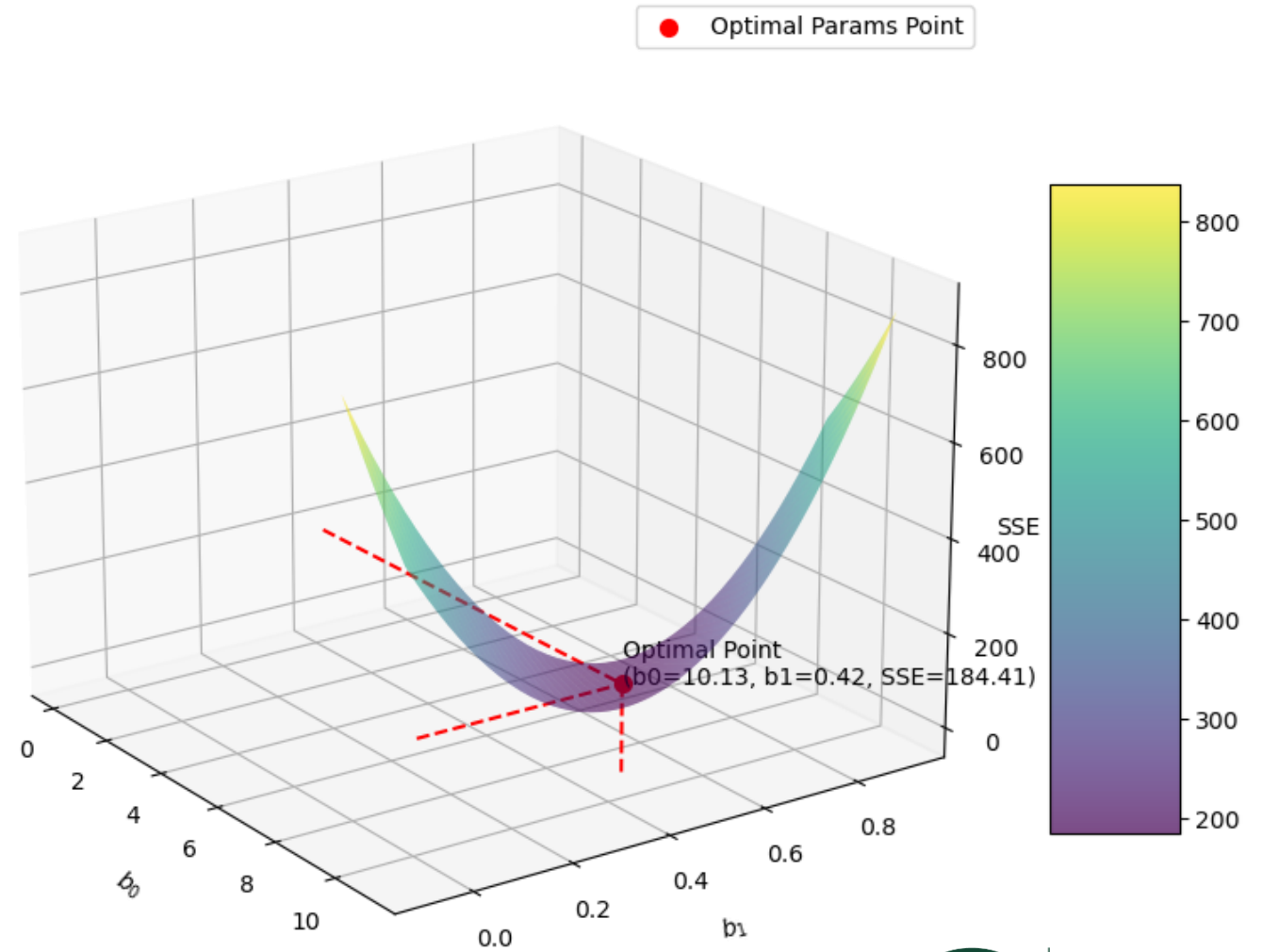
● Optimal Params Point



$p\text{-val}_{\text{start}}=0.81; p\text{-val}_{\text{end}}=0.49$

Simultaneous Testing

SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Simultaneous Testing



$p\text{-val}_{\text{start}}=0.99; p\text{-val}_{\text{end}}=0.99$

Can we use the Optimization Model to find a 'Good' Transformation?

$$\text{minimize } \sum_i f_i(\epsilon_i) = \epsilon_i^2 = \left(Y_i - (\beta X_i^k) \right)^2$$

s. t.

$$p_{\text{Independence}} \geq p_{\text{Independence}}^*$$

$$p_{\text{Normality}} \geq p_{\text{Normality}}^*$$

$$p_{\text{Fit}} \geq p_{\text{Fit}}^*$$

$$p_{\text{Homoskedasticity}} \geq p_{\text{Homoskedasticity}}^*$$



Minimizing Sum of Absolute Errors

Problem Formulation

$$\text{minimize } \sum_i f_i(\epsilon_i) = |\epsilon_i| = |Y_i - (\beta_0 + \beta_1 X_i)|$$

s. t.

$$p_{\text{Independence}} \geq p_{\text{Independence}}^*$$

$$p_{\text{Normality}} \geq p_{\text{Normality}}^*$$

$$p_{\text{Fit}} \geq p_{\text{Fit}}^*$$

$$p_{\text{Homoskedasticity}} \geq p_{\text{Homoskedasticity}}^*$$

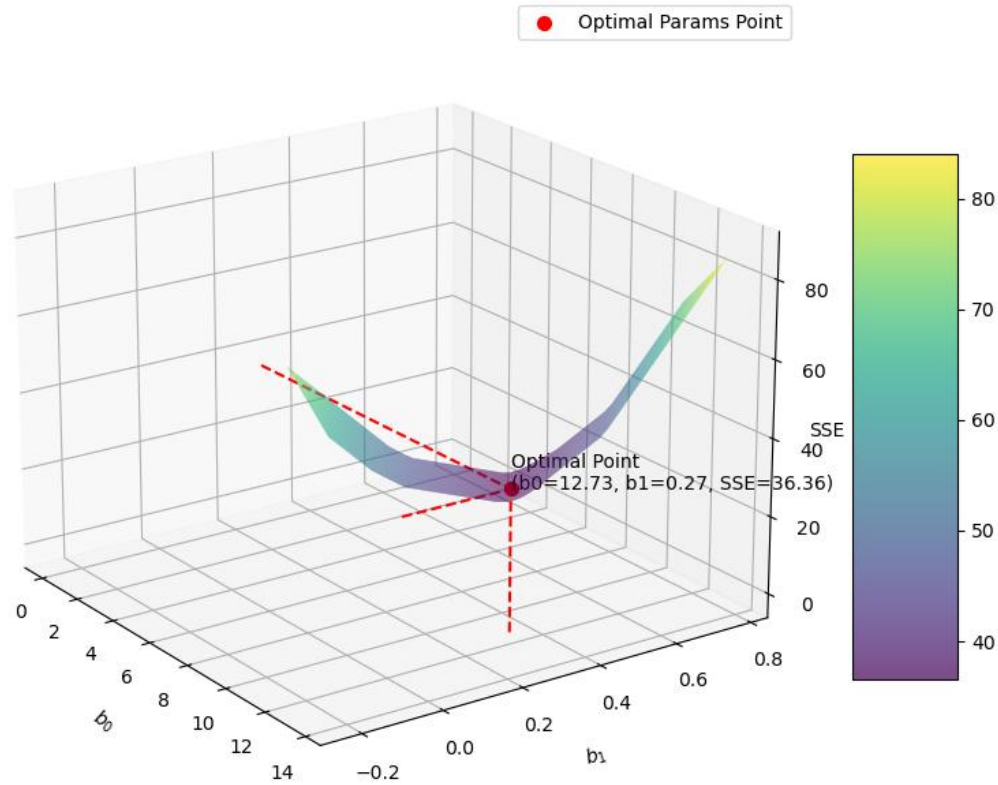
$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta} \end{pmatrix} = \arg \min (||y - (\beta_0 + X\beta)||_1)$$

s. t.

$G - M$ constraints

Independence

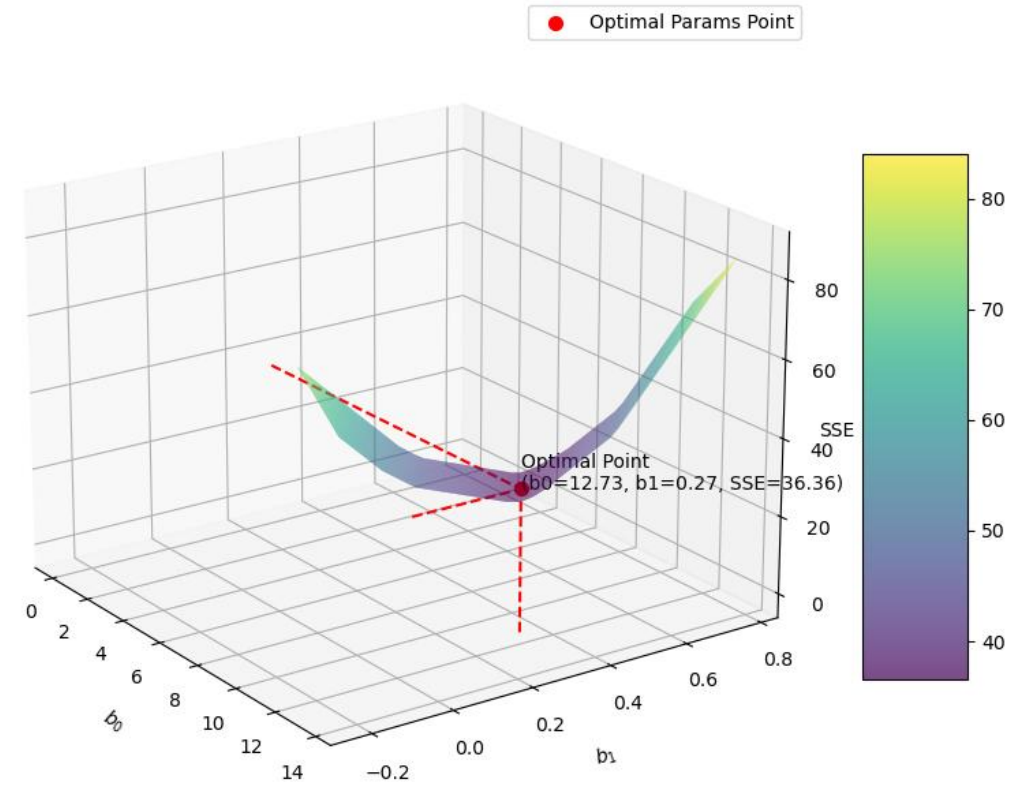
SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Independence Testing



$p\text{-val}_{\text{start}}=0.11$; $p\text{-val}_{\text{end}}=0.45$

Normality

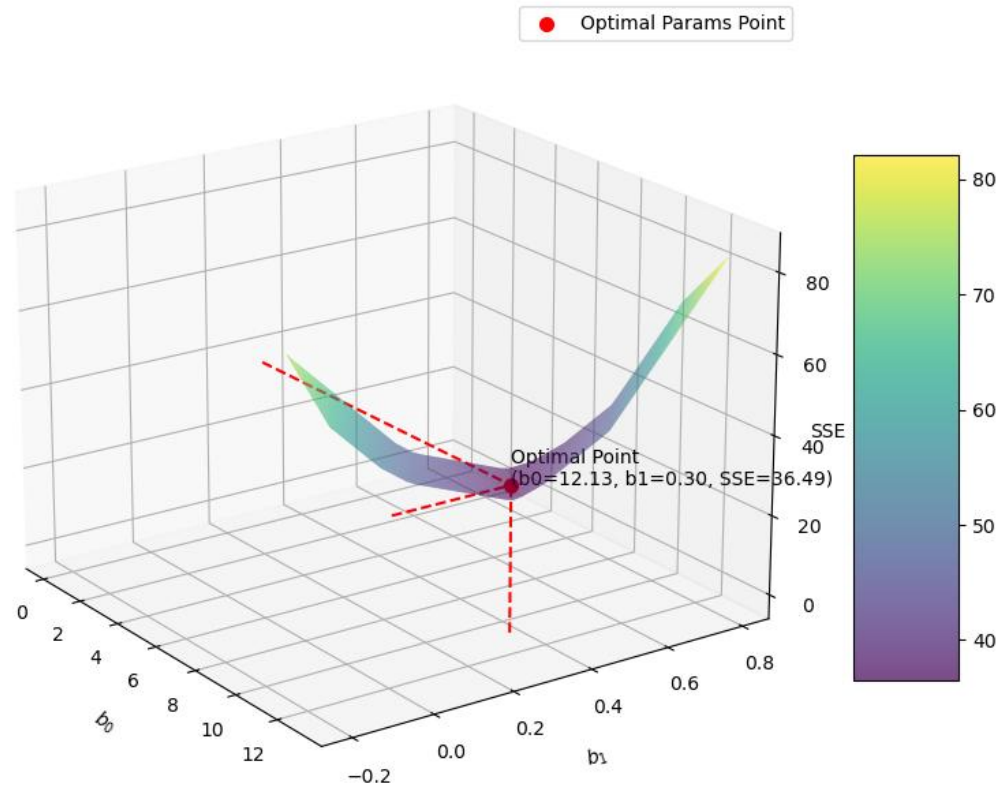
SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Normality Testing



$p\text{-val}_{\text{start}}=0.07$; $p\text{-val}_{\text{end}}=0.62$

Fit/Linearity

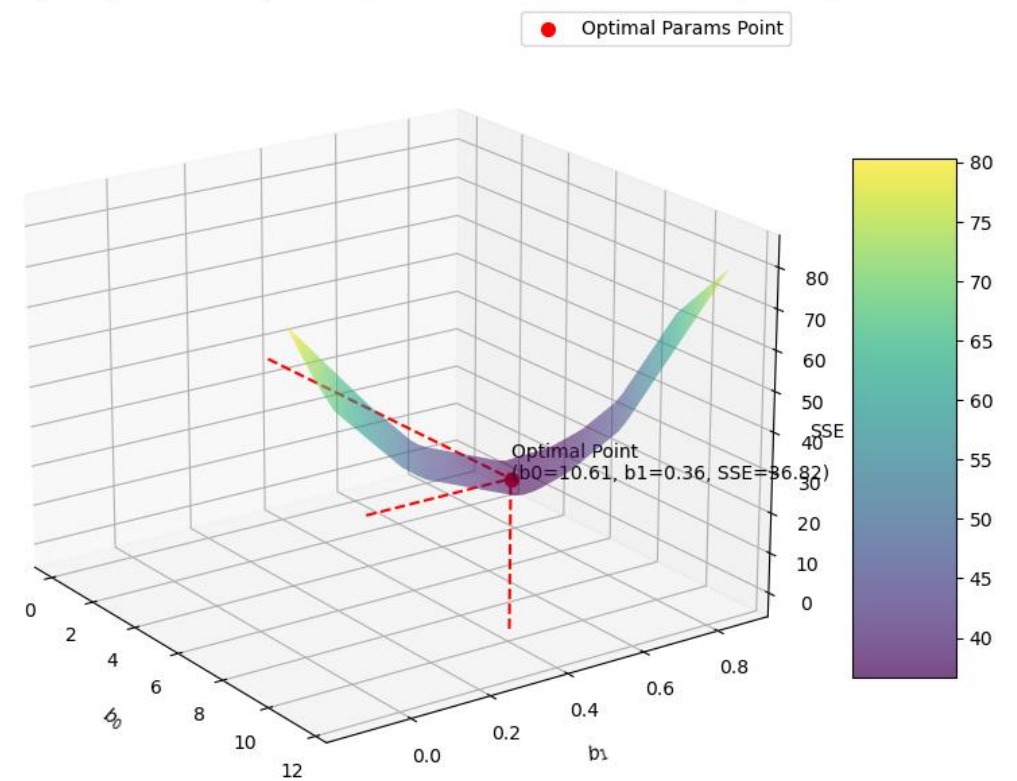
SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Linearity Testing



$p\text{-val}_{\text{start}}=0.99$; $p\text{-val}_{\text{end}}=0.99$

Homoscedasticity

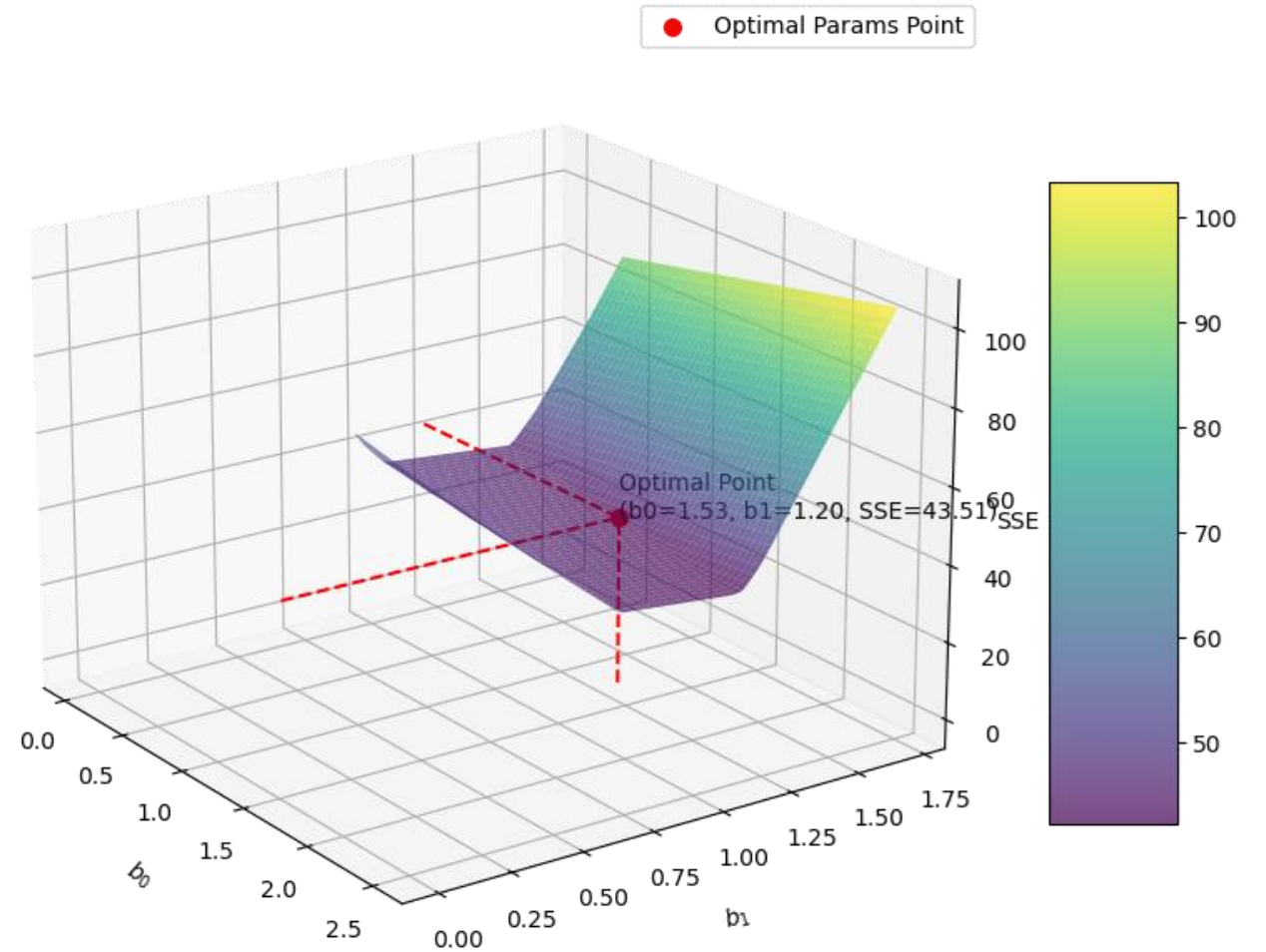
SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Heteroscedasticity Testing



$p\text{-val}_{\text{start}}=0.81$; $p\text{-val}_{\text{end}}=0.44$

Simultaneous Testing

SSE vs. b_0 vs. b_1 Parameter Space Response Surface: Simultaneous Testing



$p\text{-val}_{\text{start}}=0.99$; $p\text{-val}_{\text{end}}=0.95$